

Key points

- Fungi are eukaryotic, with rigid cell walls composed mainly of fibrillar polysaccharides, largely β -1,3-glucan and chitin, that bind covalently to each other and to proteins. The fungal cell membrane contains ergosterol rather than the cholesterol, as in mammalian cells.
- Fungal diseases are called mycoses and are often chronic infections. They may involve only the skin extending into the epidermis or may cause subcutaneous or systemic infections.
- Clinical response to antifungal treatment can be lengthy because fungi are complex slow-growing organisms with thick walls often embedded deep in tissues.
- Patients with chronic immune suppression due to organ transplantation, cancer chemotherapy, or HIV infection are susceptible to opportunistic fungal infections. Morbidity is high in immunosuppressed patients.
- Targets of antifungal drugs include components of the cell wall, ergosterol in the membrane, and nuclear functions.
- Drugs for the treatment of systemic mycoses include the polyene antifungals, azole antifungals, and echinocandins.
- Many azole antifungals and other structurally diverse drugs are used topically for superficial fungal infections.

Polyenes: Amphotericin B and Nystatin: Amphipathic amphoteric molecules

Bind ergosterol → form pores in the membranes → leakage

Broad spectrum, Resistance is rare.

- Amphotericin B: I.V. | life-threatening infections | nephrotoxicity, electrolyte wasting, infusion rxns
- Lipid / liposomal formulations reduce toxicity.
- Nystatin: topical only due to severe toxicity, including nephrotoxicity, when given IV

Flucytosine → 5-FU: Inhibits fungal DNA and RNA synthesis → leads to impairment of protein synthesis and fungal cell proliferation

- Resistance: ↓ uptake by cytosine permease, ↓ conversion to active nucleoside analog by cytosine deaminase, ↓ conversion to toxic nucleotides, upregulated pyrimidine synthesis
- Narrow spectrum | oral only, high concentrations in CNS | always used in combination with another antifungal | bone marrow toxicity

Azoles (ketoconazole, itraconazole, fluconazole, voriconazole, isavuconazole, posaconazole)

- Inhibit lanosterol 14- α -demethylase → ↓ ergosterol synthesis in cell membrane → reduces membrane permeability, structural integrity, and impairs function
- Resistance: ERG11 mutations, efflux pumps
- Broad spectrum (in general) | topical treatment of superficial infection, oral/IV systemic infections
- ↑ serum liver enzymes/hepatotoxicity, QT prolongation, CYP interactions

Key points: Targets and Drug Classes

Echinocandins (caspofungin, micafungin, anadulafungin): Inhibit β -1,3-D-glucan $\rightarrow$ weakened cell wall $\rightarrow$ osmotic influx of water $\rightarrow$ cell swells $\rightarrow$ cell ruptures (lysis)

- Narrow spectrum: *Candida* spp, *Aspergillus* | IV, poor CNS penetration | low toxicity, few DDIs

Terbinafine: Inhibits squalene epoxidase (an early step in ergosterol synthesis) $\rightarrow$ reduces membrane permeability, structural integrity, and impairs function

- Resistance is rising: mutations in squalene epoxidase gene, efflux pumps
- Topical use for tinea, cutaneous candidiasis | systemically for fungal nail infections (onychomycosis)
- GI discomfort, headache, hepatotoxicity, rash | CYP2D6 inhibitor, moderate potential for drug interactions

Griseofulvin: Binds microtubules in keratin precursor cells $\rightarrow$ disrupts the mitotic spindle $\rightarrow$ inhibits fungal cell mitosis

- Resistance: altered microtubule structure, decreased uptake of drug
- Narrow spectrum: tinea infections only (skin, hair, nails)
- Oral | | long duration of treatment | GI upset, headache, $\uparrow$ serum liver enzymes | contraindicated in pregnancy

Clinical response to antifungal treatment can be protracted because fungi are complex slow-growing organisms with thick walls often embedded in tissues.

Opportunistic fungal pathogens

- *Candida* spp.
 - *Aspergillus* spp.
 - *Cryptococcus neoformans*
 - *Pneumocystis jirovecii*
 - Mucorales order
 - *Rhizopus*
 - *Mucor*
 - *Rhizomucor*
 - and other species
- The most common causes of human infections in this group

Endemic fungal pathogens

- *Blastomyces*: Midwest, south-central, southeastern regions
- *Coccidioides*: Southwestern US
- *Histoplasma*: Ohio and Mississippi River valleys
- *Paracoccidioides*: South America
- *Sporothrix*: worldwide
- others

- ❖ Opportunistic fungi are usually residents of the human microbiota or saprophytes in the environment. *Candida* and *Aspergillus* are the most common infections. Immunodeficient patients are at highest risk of opportunistic infection.
- ❖ Endemic fungi are not part of the normal human microbiome. Infection can result from inhalation of spores. (*Sporothrix* is through skin contact typically.) People who live in or travel to geographic areas where certain fungi are endemic are at risk for these infections. Weakened immune system or chronic lung disease increases the risk of endemic infection.
- ❖ The incidence of fungal infections is rising. Morbidity in immunocompromised patients is high.

Example of Antifungal Treatment Regimen: Treatment of Cryptococcal Meningitis

WHAT YOU NEED TO KNOW:

Induction, Consolidation, Maintenance,
the drugs being used and why

Amphotericin B: First-line in combination with flucytosine (patients with or without AIDS)

INDUCTION phase	Liposomal AMB or AMB lipid complex I.V. 1x daily PLUS Flucytosine (oral) q6h x 2-4 wks or until afebrile and CSF cultures are negative – Monitor / manage intracranial pressure – For HIV/AIDS patients, deferring ART for 5 weeks after initiation of therapy significantly improved survival compared to starting ART during the first 2 weeks. ART=Antiretroviral therapy
CONSOLIDATION phase	Fluconazole 400-800 mg 1x daily for 8 weeks (non-AIDS) / 10 weeks (AIDS)
MAINTENANCE phase (suppression)	Fluconazole 200mg 1x daily to prevent relapse 6-12 months (non-AIDS) – For HIV/AIDS patients, discontinuation of therapy may be considered in patients who remain asymptomatic, with CD4 cell count >100 cells/mm³ on ART

Check your knowledge

What are the advantages of liposomal AMB and AMB lipid complex compared to AMB deoxycholate?

What is AMB's mechanism of antifungal action?

What are the toxicities associated with AMB administration?

Check your knowledge

What physical properties of the amphotericin B molecule are probably important for its fungus-killing activity?

What accounts for its relatively selective toxicity against fungi?

What are the likely mechanisms of resistance?

AMB is an important therapeutic for what patients and infections?

Flucytosine (5-FC) Pharmacokinetics

PK	Oral (prodrug); rapid, good absorption, bioavailability ~90%, good CSF penetration , intracellular deamination in yeasts and certain gut bacteria to active 5-FU; urine excretion of 5-FC (renal dosing adjustments are necessary); $t_{1/2}$ 3-6h, up to 200h in renal failure; admin q6h	
MOA	Inhibits RNA and DNA synthesis (<i>See next slide.</i>) Resistance emerges rapidly during therapy when the drug is used alone. • Loss of cytosine permease transporter • ↓ Cytosine deaminase activity or uPRT activity	
Activity	Fungistatic Narrow spectrum: <i>Cryptococcus neoformans</i> , <i>Candida</i> some species, most fungi that cause chromoblastomycosis, and some dematiaceous (darkly pigmented) fungi	• AMB is nephrotoxic in many patients. • 5-FC dose must be adjusted in renal insufficiency. • 5-FC toxicities are concentration-dependent.
Uses	<i>Cryptococcus</i> meningitis and selected invasive candidiasis 5-FC has additive or synergistic effect with AMB (in vitro testing).	
AE	Well tolerated when renal function is normal Concentration-dependent adverse effects: • Heme: leukopenia, thrombocytopenia • Liver: elevated liver function enzymes (AST/ALT); liver necrosis reported • GI: nausea, vomiting diarrhea Pregnancy: adverse events seen in animal reproduction studies	Certain bacteria in the gut microbiome convert 5-FC to 5-FU, which is absorbed and can cause host cell toxicity.

Check your knowledge

What is the basis of flucytosine's selectivity?

What is the basis of flucytosine's toxicity?

How does flucytosine induce cellular cytotoxicity?

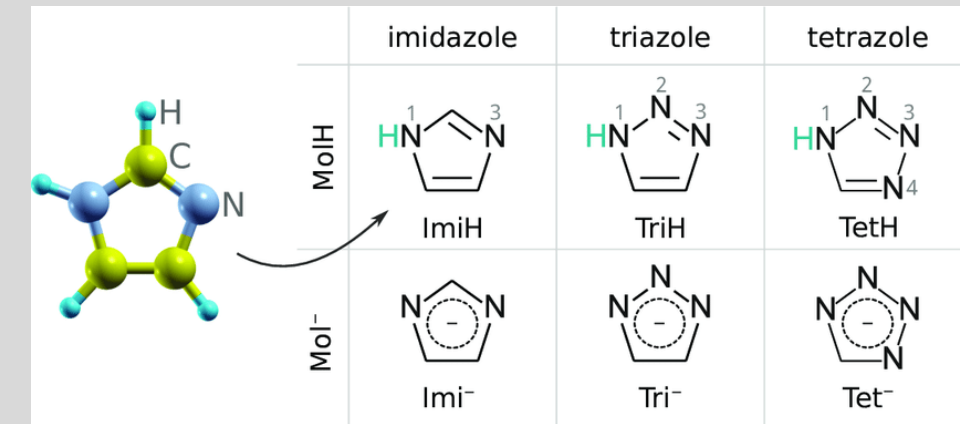
What are the therapeutic uses of 5-FC and why?

What are major adverse effects of 5-FC and how can they be reduced?

Azole Antifungal Agents

Broad spectrum of activity | Fungistatic

Some azoles appear to fungicidal against *Aspergillus* spp.



For Systemic Use

Triazoles

- *Itraconazole
- *Fluconazole
- *Voriconazole
- *Posiconazole
- *Isavuconazole (Prodrug: Isavuconazonium)

Oteseconazole

Oteseconazole is a tetrazole antifungal drug for oral treatment of recurrent vulvovaginal candidiasis.

*Ketoconazole

Because of its toxicity, oral ketoconazole has been replaced by systemic triazole antifungals.

Topical Products

- *Clotrimazole
- *Ketoconazole
- Butoconazole
- Miconazole
- Econazole
- Luliconazole
- Oxiconazole
- Sertaconazole
- Sulconazole
- Tiaconazole
- Terconazole
- Efinaconazole

Imidazoles

Pharmacokinetics of Azole Antifungal Agents

TAKEAWAYS: Low protein binding → diffusion across BBB. Bioavailability and $t_{1/2}$ vary. Extensive hepatic metabolism (most drugs). Inhibition of CYP3A4 (and others).

	Route Bioavailability	CSF Penetration	Elimination / substrate CYP inhibition	$t_{1/2}$ (hours)	Drug Interactions
Itraconazole	Oral Variable*	No	Hepatic / CYP3A4 CYP3A4, P-gp	24-42	High potential for drug interactions
Fluconazole	Oral, IV >90%	Yes	Renal / minor hepatic ~80% unchanged (urine) CYP2C19, 2C9, 3A4	22-31	
Voriconazole	Oral, IV 96%	Yes	Hepatic CYP2C19, 2C9, 3A4 CYP3A4, 2C9	6	
Posaconazole	Oral, IV Variable*	No	Minimal (~17% by UGT) excreted unchanged (feces) CYP3A4, P-gp	25	
Isavuconazole	Oral, IV 98%	limited data	Hepatic / CYP3A4, UGT CYP3A4, P-gp	130	
Ketoconazole	Oral Variable*	No	Hepatic CYP3A4, P-gp	7-10	

*Itraconazole, Posaconazole, Ketoconazole: acid environment for optimal absorption

Bold: strong inhibitors

Summary of azole antifungals spectrums of activity (susceptible fungi)

	Itraconazole	Fluconazole	Voriconazole	Posaconazole	Isavuconazole
Aspergillosus	✓	—	✓	✓	✓
Candidiasis	✓	✓	✓	✓	✓
C. neoformans	✓	✓	✓	✓	✓
Mucormycosis	—	—	—	✓	✓
Dimorphic fungi	✓	±	✓	**	**
Onychomycosis	✓	±	**	**	**
Tinea	✓	✓	**	**	**
— No activity ± Less effective for these fungal infections than other options. ** Effective but reserved for infections that can't be treated with standard therapy. Tinea = Dermatophytosis					

TAKEAWAY: The spectrum of action directly influences usefulness in treatment specific infections. Remember to think about risks of resistance and adverse effects for the individual patient.

Azole Antifungals Therapeutic Uses (susceptible fungi)

TAKEAWAYS:
Spectrums of action→uses, drug of choice, CNS penetration

Itraconazole	Fluconazole	Voriconazole	Posaconazole	Isavuconazole
Broad spectrum	Good CNS penetration	Good CNS penetration	Broadest spectrums of the azole antifungals. Generally reserved for patients refractory to or intolerant of first-line drugs.	
Drug or choice for dimorphic fungi • Blastomycosis • Sporotrichosis • Histoplasmosis • Coccidiomycosis • Paracoccidioidomycosis Alternative for: • Onychomycosis (second to terbinafine) • Tinea infections refractory to topical antifungal treatment • Aspergillosis • Candidiasis	Initial choice for • Cryptococcal meningitis • consolidation and maintenance phases • Candidemia • Mucocutaneous candidiasis • Coccidioidomycosis No activity against aspergillosis or mucormycosis	Drug of choice for invasive infections • Aspergillosis • Candidemia, including fluconazole-resistant • Disseminated <i>Candida</i> infections • Cryptococcal meningitis (alternative for fluconazole) • Serious infections by dimorphic fungi	Treatment of invasive infections • Aspergillosis • Mucormycosis, oral step down therapy • Oropharyngeal candidiasis • Salvage therapy for invasive fungal infections	Treatment of invasive infections • Aspergillosis • Mucormycosis Alternative for • Cryptococcal meningitis (maintenance and consolidation) • ISA penetrates brain parenchyma. • Esophageal candidiasis

Posaconazole and isavuconazole are the only azoles with in vitro activity against Mucorales.

Itraconazole, voriconazole, posaconazole and isavuconazole are also used for prophylaxis of invasive fungal infections in hematologic malignancy, hematopoietic stem cell transplant, and solid organ transplant patients.

Azole antifungals: Class adverse effects

Monitor for signs and symptoms; Monitor blood levels as appropriate.

System	Class effects unless otherwise indicated (not a complete list)	
GI	The most common side effect: Nausea, abdominal pain	
Hepatotoxicity	Ranges from asymptomatic increase in serum hepatic enzymes (AST/ALT) to liver failure (rare)	
Cardiac	Prolongation of QT interval → torsades de pointes → ventricular arrhythmias.	
Neurological	Peripheral neuropathy Onset: 1 to 18 months	Most common with itraconazole Also occurs with voriconazole and posaconazole
Hypersensitivity reactions	Immediate: urticaria, angioedema, anaphylactic shock Delayed: Range from rash to severe cutaneous reactions (SJS/TENS) Serum sickness, and others Photosensitivity	
Pregnancy	Avoid – may cause fetal harm. Use only if benefit to mother outweighs risk to fetus.	
Drug interactions <i>High potential</i>	CYP3A4-related: Co-administration with many drugs metabolized by CYP3A4 is contraindicated. QT interval prolongation: Drugs that prolong QT interval or patients with long QT syndrome may cause torsades de pointes (ventricular arrhythmias) pH-related: Ketoconazole, itraconazole caps, posaconazole require acidic environment (with food or acidic carbonated beverage, eg, ginger ale); some should be taken on an empty stomach.	

Voriconazole and posaconazole IV formulated with cyclodextrin can accumulate in renal insufficiency and acute kidney injury.

Itraconazole	Congestive heart failure	Itraconazole is contraindicated in patients with heart failure.
Voriconazole	- Ophthalmic effects: Blurred vision, optic neuritis, vision color changes, scleritis - Peripheral neuropathy reported: May be rapid, debilitating, irreversible - Encephalopathy, hallucinations may occur with IV administration (high concentrations) - Periosteal disease: Voriconazole is trifluorinated; F⁻ is integrated into extracellular matrix.	
Metabolic-related effects: Proposed mechanisms		
Adrenal suppression		
Ketoconazole: strong inhibitor of steroidogenesis Itraconazole and posaconazole: Effects on steroidogenesis are milder and occur less frequently.		
Inhibition of 17-alpha hydroxylase (CYP17) and C17,20 desmolase suppresses steroidogenesis.		
Suppression of androgen synthesis leads to oligospermia, gynecomastia, menstrual irregularities (estrogens are derived from androgens), and decreased libido.		
Inhibition of 11-β hydroxylase (CYP11A1) decreases glucocorticoid and mineralocorticoid synthesis.		
Cortisol and aldosterone synthesis are suppressed		
Pseudohyperaldosteronism with hypertension and hypokalemia:		
Itraconazole and posaconazole: More frequently observed		
Azoles may inhibit the enzyme that inactivates cortisol, 11-β-hydroxysteroid dehydrogenase type 2, which is found in aldosterone-sensitive tissues.		
Like aldosterone, cortisol activates mineralocorticoid receptors → sodium and water reabsorption, which raises blood pressure, and causes potassium loss.		
Fluconazole, voriconazole, isavuconazole: Low potential for adrenal suppression and pseudohyperaldosteronism.		

- **When prescribing oral azole antifungal therapy, certain precautions should be considered and recommendations should be made to the patient.**
 1. The patient's concomitant drugs and comorbid conditions in view of the high potential for drug interactions.
 2. The patient's risk of ventricular fibrillation in view of the drug's QT interval prolonging effects.
 3. Food?: Take with food (for gastric acidity) or take on empty stomach?
 4. Patient education about the symptoms and signs of potential side effects (class and drug-specific effects) and hypersensitivity reactions.
 5. The importance of monitoring: liver function studies and therapeutic drug levels as appropriate.

Check your knowledge

Which drug is first line for treatment of fungal infections by dimorphic fungi?

Which drug is first line for consolidation and maintenance of cryptococcal meningitis?

Which drug is the preferred initial treatment for invasive aspergillosis?

Which drug is an alternative for the initial treatment of invasive aspergillosis in a patient with optic neuritis or peripheral neuropathy?

Which drugs are used as step-down therapy for mucormycosis?

Which oral drug is first line for vulvovaginal candidiasis?

Check your knowledge

What adverse effects can occur with all of the azole antifungals?

Which azole antifungal should be avoided in patients with a history of heart failure?

Which azole antifungal is associated with optic neuritis and other ocular toxicities?

Which azole antifungal is associated with peripheral neuropathy?

Which azole antifungal is most likely to interfere with steroidogenesis?

Check your knowledge

What is the major mechanism of drug interactions with the azole antifungal drugs?

One of the azole antifungals is cleared renally and excreted unchanged in urine. Which one? What do we need to think about in terms of dosing this drug?

Two triazole antifungal drugs are formulated with cyclodextrin to solubilize the hydrophobic drugs for intravenous administration. What toxicity can occur in patients with renal insufficiency? What are the two drugs?

One of the antifungals in this group does not require dosage adjustment for patients with renal insufficiency or liver impairment. What is it? Why are dosage adjustments not required?

Caspofungin Therapeutic Uses: **Serious Infections**

Drug	Candidemia (fungicidal)	Invasive Aspergillosis (fungistatic)	Neutropenic fever	Prophylaxis*
Caspofungin adults / peds	✓	✓	✓	✓
Micafungin adults / peds	✓	✓	✓	✓
Anidulafungin adults / peds	✓	✓	✓	✓
Rezafungin adults	✓	effective but not approved		

*Prophylaxis against invasive fungal infections

Patients at risk for invasive fungal infections include acute leukemia (neutropenia and intensive induction therapy) and hematopoietic stem cell transplantation.

Adverse Effects

Generally well tolerated

diarrhea, nausea, abdominal pain

- Phlebitis at infusion site
- Histamine-mediated effects: flushing, urticaria, hypotension
- Hepatic impairment: ↑LFTs; hepatitis reported.
- Cardiotoxicity – low rates reported

Pregnancy: May cause fetal harm, based on animal reproduction studies: weigh risk/benefit

Drug interactions: Monitor therapy

- Caspofungin + Cyclosporine → may ↑ each other's levels → may ↑ toxicity of each
- Rifampin and other inducers of clearance may increase caspofungin levels.
- Micafungin + Sirolimus → ↑sirolimus levels
- All of them may decrease therapeutic effects of *Saccharomyces boulardii* probiotic yeast supplement for tx of antibiotic-associated *C. difficile* diarrhea.

Check your knowledge

What is the fungal cell target of the echinocandins?

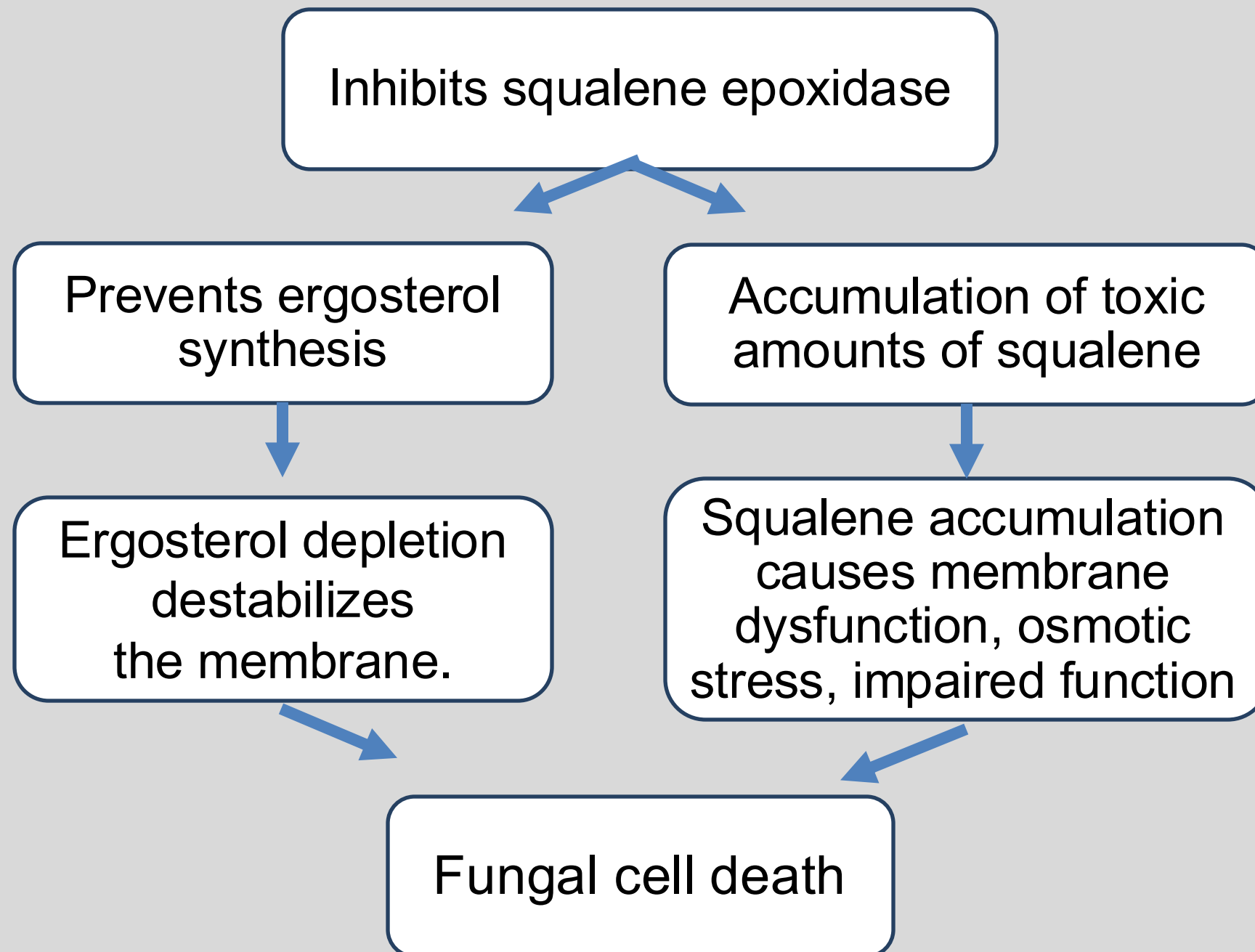
What is the spectrum of action of echinocandins?

What is the clinical importance of the echinocandins?

What is the effect of drug interactions with concurrently administered drugs?

What are the recommendations for use of glucan synthase inhibitors in pregnancy?

Terbinafine accumulates in keratin —Fungicidal—



Terbinafine Treatment

Spectrum: Tinea and *Candida*

Drug of choice for

- onychomycosis (oral)
- (more effective than griseofulvin and itraconazole for fungal infections of fingernails and toenails)

Treatment x3 months

- dermatophytoses (topical) and
- cutaneous candidiasis (topical)

Review: Compare and contrast griseofulvin and terbinafine.

ADME
Mechanism of action
Therapeutic uses
Adverse effects
Drug interactions
Contraindications

Drug	Mechanism of action	Spectrum of activity
*Nystatin – Polyene	Binds ergosterol forming pores in the plasma membrane	<i>Candida</i> only
*Azoles	14- α -demethylase inhibitors	Fungistatic <i>C. albicans</i> , Dermatophytes, <i>Malassezia furfur</i> (pityriasis versicolor)
Allylamines: *Terbinafine, Naftafine Benzylamine: Butenafine	Squalene epoxidase inhibitors	Fungicidal Dermatophytes <i>Malassezia furfur</i>
Ciclopirox	Chelates trivalent metal ions → inhibits metal-dependent enzymes; also blocks intracellular transport, which disrupts DNA, RNA, protein synthesis	Fungicidal <i>C. albicans</i> , Dermatophytes, <i>Malassezia furfur</i>
Tolnaftate	Distorts the hyphae and stunts growth of susceptible fungi; may involve inhibition of squalene epoxidase	Fungicidal Dermatophytes <i>Malassezia furfur</i>
Tavaborole	Inhibits leucyl-tRNA synthase → protein synthesis inhibitor	Fungistatic Trichophyton

Mycotic infections may involve the skin or subcutaneous tissues or may cause systemic infections.

Fungi are eukaryotic with rigid cell walls composed of chitin. Their membrane contains ergosterol, rather than cholesterol. These structures are useful targets of drug therapy.

Amphotericin B (AMB):

- Broad spectrum of activity; fungicidal
- AMB is a polyene antibiotic binds ergosterol in fungal cell membrane causing pore formation and leakage of cell components.
- Relative selectivity: Ergosterol has greater affinity for AMB than cholesterol has.
- Resistance is rare; possibly due to mutations in ERG gene or replacement of ergosterol with precursor sterols.
- AMB is used for the treatment of serious systemic and CNS fungal infections, including during pregnancy.
- Nephrotoxicity and infusion reactions (fever, chills, tachypnea) limit its use.

Flucytosine (5-FC)

- Prodrug converted to cytotoxic antimetabolite, 5-fluorouracil (5-FU), which is activated by phosphorylation and inhibits fungal RNA and DNA synthesis.
- Narrow spectrum of activity; fungistatic
- Resistance develops rapidly when it is used as a single agent, probably by reduced cytosine permease or cytosine deaminase activity.
- 5-FC is an adjunct in the treatment of cryptococcal meningitis, and some invasive candidiasis
- Generally well tolerated. Concentration-dependent leukopenia, nausea, vomiting, diarrhea. Monitor renal function.

Azole antifungal agents: imidazoles and triazoles.

- Broad spectrum of activity, in general; fungistatic
- PK: Most are substrates and strong inhibitors of CYP3A4 and other CYPs.
- Mechanism: Competitive inhibitors of 14- α -demethylase (CYP51), blocking the conversion of lanosterol to ergosterol leading to membrane instability and inhibits fungal cell growth.
- Resistance: *ERG11* gene mutation, leading to decreased binding affinity of 14- α -demethylase and decreased azole efficacy. Some fungal strains upregulate efflux pumps or increase production of lanosterol.
- Triazoles are generally used for treatment of systemic and cutaneous mycoses, while imidazoles are topical agents for cutaneous infections.
- QT interval prolongation is a potentially serious adverse effect of all azole antifungals. Other side effects include GI disturbances and rash. There is a high risk of drug interactions.

Echinocandins

- Narrow spectrum; *Candida* spp (fungicidal), *Aspergillus* spp (fungistatic)
- Mechanism: Noncompetitive inhibition of β -1,3-glucan synthase, which decreases cell wall synthesis.
- Resistance occurs by FKS-1 and FKS-2 alterations that result in decreased affinity for the echinocandins.
- The drugs are well tolerated. Flushing may occur. Caspofungin increases systemic concentrations of the immunosuppressants cyclosporine and tacrolimus, and micafungin increases sirolimus levels. Cautious use of the echinocandins is recommended in patients taking these immunosuppressants.

Terbinafine

- Broad spectrum, fungicidal
- Mechanism: Noncompetitive inhibition of squalene epoxidase, which prevent ergosterol biosynthesis, leading to cell death.
- Resistance is reported and linked to mutations in squalene epoxidase gene, SQLE
- Terbinafine accumulates in keratin-rich tissues and is more effective than griseofulvin in the treatment of dermatophytoses of the nails, skin, and scalp; available for oral and topical use.
- Side effects include GI upset and headache. Terbinafine it is contraindicated in patients with chronic or active hepatic disease. Terbinafine is a CYP substrate and CYP2D6 inhibitor.

Griseofulvin

- Narrow spectrum – dermatophytes; fungistatic
- Binds human keratin in skin, hair, and nails and causes disruption of the mitotic spindle and inhibition of mitosis
- Treatment of dermytophytoses of the scalp and hair. It has been largely replaced by terbinafine for the treatment of onychomycosis. Long duration of therapy – 6-18 months.
- Ultramicrosize griseofulvin is adequately absorbed and absorption is enhanced by fatty food.
- Side effects include GI upset, headache, rash, and photosensitivity, It is contraindicated in pregnancy and patients with porphyria or hepatic failure.

Topical antifungal agents

- A number of antifungal drugs are used topically for superficial infections caused by *C. albicans* and dermatophytes.
- Preparations include oral suspension, oral troches, vaginal creams and suppositories, skin creams, ointments, and powder.

Nystatin

- Polyene antibiotic that disrupts fungal membranes by binding to ergosterol.
- (Recall: Amphotericin B is a polyene antibiotic.)
- Topically to suppress local *Candida* infections, including mild oral-pharyngeal candidiasis (thrush)
- Orally for secondary prevention of fungal peritonitis in patients on peritoneal dialysis
- Nystatin is poorly absorbed from the gastrointestinal tract.
- Nephrotoxic (like AMB), therefore nystatin is not administered systemically.

Azole compounds:

- miconazole, clotrimazole, ketoconazole, and several others and terbinafine topical products are used for superficial tinea infections and mucocutaneous *Candida* infections.

Terbinafine, naftifine, butenafine: Squalene epoxidase inhibitors topically for tinea infection and cutaneous candidiasis

Tolnaftate: Distorts hyphae and stunts fungal growth for treatment of tinea infections

Ciclopirox: Inhibits transport of essential molecules and interferes with enzyme functions, DNA repair, and cell division signals and structures.

Tavaborole: tRNA synthase inhibitor for treatment of onychomycosis.

Check your knowledge

What are the advantages of liposomal AMB and AMB lipid complex compared to AMB deoxycholate?

- ***Infusion-related toxicity and nephrotoxicity are reduced***, which permits use of larger doses compared to conventional AMB.
- AMB lipid formulations enhance delivery to the site of infection while reducing organ toxicity.
- Higher doses can improve effectiveness, particularly in severe or resistant fungal infections.

What is AMB's mechanism of antifungal action?

- AMB associates with ergosterol in fungal membrane causing pores to form. Cations and other small molecules leak out.
- Ergosterol is sequestered further compromising membrane integrity, leading to fungal cell death.

What are the toxicities associated with AMB administration?

- Infusion reactions, nephrotoxicity (azotemia, acute kidney injury, renal tubular acidosis), and anemia (impaired EPO synthesis)

Check your knowledge

What physical properties of the amphotericin B molecule are probably important for its fungus-killing activity?

- The amphipathic property allows the drug to interact with both hydrophilic polar head groups of the cell membrane and hydrophobic membrane core and ergosterol embedded in the membrane.
- The amphoteric property (acid and base functional groups) enhances binding to ergosterol and helps to sequester ergosterol, further destabilizing the cell membrane.

What accounts for its relatively selective toxicity against fungi?

- AMB affinity for ergosterol in fungal membranes is many times greater than its affinity for cholesterol in mammalian membranes.
- What are the likely mechanisms of resistance?
- Reduced concentrations of ergosterol due to mutations in various ERG genes or replacement with sterol precursors. Resistance to AMB is rare.

AMB is an important therapeutic for what patients and infections?

- AMB is primarily for treatment of patients with ***progressive and potentially life-threatening fungal infections***.
- AMB should ***not*** be used to treat noninvasive forms of fungal disease such as oral thrush, vaginal candidiasis, and esophageal candidiasis in patients with normal neutrophil counts.

Check your knowledge

What is the basis of flucytosine's selectivity?

- Mammalian cells cannot activate 5-FC as they do not have cytosine permease or cytosine deaminase.

What is the basis of flucytosine's toxicity?

- Intracellular conversion to toxic 5-FU in fungal cell

How does flucytosine induce cellular cytotoxicity?

- 5-FUTP incorporation of FUTP into RNA → ↓RNA synthesis (thus, ↓protein synthesis)
- 5-FdUMP inhibits thymidylate synthase → ↓ DNA synthesis

What are the therapeutic uses of 5-FC and why?

- Narrow spectrum of activity
- Cryptococcal meningitis with AMB and selected invasive candidiasis with AMB.
- Oral, crosses BBB; rapid development of resistance when used alone

What are major adverse effects of 5-FC and how can they be reduced?

- Well tolerated when renal function is normal – Adverse effects are concentration-dependent
- Leukopenia, thrombocytopenia / elevated liver function enzymes (AST/ALT) / nausea, vomiting diarrhea
- Monitor renal function

Check your knowledge

Which drug is first line for treatment of fungal infections by dimorphic fungi?

- Itraconazole

Which drug is first line for consolidation and maintenance of cryptococcal meningitis?

- Fluconazole
- Alternatives are voriconazole, posaconazole and isavuconazole (isavuconozium is the prodrug).
- Initial therapy regimen is amphotericin B + flucytosine

Which drug is the preferred initial treatment for invasive aspergillosis?

- Voriconazole

Which drug is an alternative for the initial treatment of invasive aspergillosis in a patient with optic neuritis or peripheral neuropathy?

- Posaconazole or isavuconazole and they are better tolerated than voriconazole.
- Voriconazole can cause ocular, neurological, and severe dermatologic toxicities.

Which drugs are used as step-down therapy for mucormycosis?

- Posaconazole and isavuconazole
- Amphotericin B is the preferred initial therapy.

Which oral drug is first line for vulvovaginal candidiasis?

- Fluconazole

Check your knowledge

What adverse effects can occur with all of the azole antifungals?

- Minor GI disturbances (eg nausea, diarrhea); ↑ serum aminotransferases (hepatic enzymes), hepatitis (rare); prolongation of QT interval; TdP/ventricular arrhythmias; metabolic-related hypokalemia, pseudoaldosteronisms and hypertension; immediate and delayed hypersensitivity reactions

Which azole antifungal should be avoided in patients with a history of heart failure?

- Itraconazole has negative inotropic effects (reduced force of contraction) and can worsen heart failure AND it is associated with new heart failure and with new and worsening hypertension. Metabolic-related effects observed more often than with other triazoles – except posaconazole.

Which azole antifungal is associated with optic neuritis and other ocular toxicities?

- Voriconazole: Optic neuritis, scleritis, episcleritis, photophobia

Which azole antifungal is associated with peripheral neuropathy?

- Voriconazole: Numbness and tingling in the extremities may be rapidly debilitating and irreversible. Encephalopathy is reported.

Which azole antifungal is most likely to interfere with steroidogenesis?

- Ketoconazole, an imidazole antifungal that is currently used only topically, has a high risk of adrenal suppression due to its strong inhibition of steroidogenesis.
- Itraconazole and posaconazole have a moderate risk of adrenal suppression.
- FYI: Cortisol, aldosterone, and the androgen precursors of sex hormones, dehydroepiandrosterone (DHEA) and DHEA sulfate are synthesized in the adrenal cortex.

Check your knowledge

What is the major mechanism of drug interactions with the azole antifungal drugs?

- CYP inhibition. All azole antifungals are strong CYP3A4 inhibitors, except fluconazole and isavuconazole, which both inhibit CYP3A4 but with moderate affinity for the enzyme.

One of the azole antifungals is cleared renally and excreted unchanged in urine. Which one? What do we need to think about in terms of dosing this drug?

- Fluconazole, 80% of the dose is excreted unchanged in the urine.
- Dosage adjustment for patients with renal insufficiency.
- Note: Fluconazole is a strong inhibitor of CYP2C19 and inhibit CYP2C9 and 3A4 with lower affinity.

Two triazole antifungal drugs are formulated with cyclodextrin to solubilize the hydrophobic drugs for intravenous administration. What toxicity can occur in patients with renal insufficiency? What are the two drugs?

- Accumulation of cyclodextrin in patients with renal insufficiency can cause toxicity to the kidney tubular cells. The drugs are voriconazole and posaconazole.

One of the antifungals in this group does not require dosage adjustment for patients with renal insufficiency or liver impairment. What is it? Why are dosage adjustments not required?

- The drug is posaconazole because it is minimally metabolized (~17% by UGT) by the liver, excreted unchanged in the feces, and is not cleared through the kidney.

Check your knowledge

What is the fungal cell target of the echinocandins?

- They inhibit β -1,3-glucan synthase complex, which disrupts the integrity of the fungal cell wall, leading to cell death.

What is the spectrum of action of echinocandins?

- *Candida* spp (fungicidal); *Aspergillus* spp (fungistatic)

What is the clinical importance of the echinocandins?

- Treatment and prophylaxis of serious *Candida* infections; treatment of invasive aspergillosis not responsive to other drugs

What is the effect of drug interactions with concurrently administered drugs?

- Caspofungin and cyclosporine may increase plasma concentrations of each other increasing the risk of hepatotoxicity (echinocandins) and nephrotoxicity (cyclosporine)
- Micafungin increases sirolimus levels increasing the risk of numerous toxicities.
- All may diminish the therapeutic effects of *Sacharomyces boulardii*, a beneficial yeast in the gut microbiome.
- Ibrexafungerp: CYP3A4 inducers (avoid) and inhibitors (caution)

What are the recommendations for use of glucan synthase inhibitors in pregnancy?

- Echinocandins: Animal reproduction studies suggest the potential for fetal harm. Use only if benefit to the mother outweighs the risk to the fetus.

Review: Compare and contrast griseofulvin and terbinafine.

ADME	Oral, once daily G: Take with fatty food	Extensive hepatic metabolism Terminal $t_{1/2}$: G: 9-24 h; T: 200-400 h at steady state
Mechanism of action	G: binds microtubules in new keratin and prevents mitosis T: squalene epoxidase inhibitor reducing ergosterol synthesis	
Therapeutic uses	Dermatophyte infections G: DOC for tinea capitis in children T: DOC for onychomycosis DOC: drug of choice	
Adverse effects	Both generally well tolerated GI disturbances, hypersensitivity reactions	
Drug interactions	G: CYP inducer T: Strong CYP2D6 inhibitor	
Contraindications	G: liver failure, porphyria, pregnancy T: chronic or acute liver disease, avoid in pregnancy	